

These lecture notes are prepared by Salman Ahmad, who is both a teacher and a web developer. The purpose of this material is to make the topic clear and useful for students of Class 9 Computer Science.

Utility Programs

Utility programs are part of system software. They help keep a computer running smoothly, safely, and efficiently. They manage files, protect data, and improve system performance.

1. Disk Cleanup

What it does: Removes temporary files and unwanted data from the computer.

Real-life Scenario: Frees up storage space and helps the computer run faster.

Example: If your computer becomes slow, running Disk Cleanup can improve performance.

2. Antivirus Software

What it does: Scans the computer for viruses and harmful software and protects it in real time.

Why it is useful: Keeps the computer safe from malware and online threats.

Example: Before opening an email attachment from an unknown sender, antivirus software checks it for viruses.

3. Backup Software

What it does: Creates copies of important files and stores them safely.

Why it is useful: Helps recover files if they are deleted or lost.

Example: If you delete an important file by mistake, backup software helps you restore it.

4. File Compression Tools

What they do: Reduce file size by compressing files into formats like ZIP or RAR.

Why they are useful: Make files easier to store and send.

Example: Compressing photos into a ZIP file before sending them by email.

Device Drivers

Device drivers help the operating system communicate with hardware devices. They act like translators between the computer and its devices.

- **Printer Driver:** Helps the computer print documents correctly.
- **Graphics Driver:** Helps display images and videos properly.

How Device Drivers work

- **Installation:** Driver is installed when a new device is connected.
- **Communication:** Driver translates commands for the device.
- **Operation:** Device and computer work smoothly together.

Example:

A device driver is like a TV remote that sends instructions to the TV.

Application Software

Application software helps users perform specific tasks like writing, calculating, designing, or entertainment.

Commonly Used Application softwares

1. Word Processing Software
2. Spreadsheet Software
3. Graphic Design Software

Word Processing Softwares

Word processing software is used to create, edit, format, and print text documents such as letters, reports, essays, and assignments.

Examples of Word Processing Software

Microsoft Word

Developer: Microsoft

It is one of the most commonly used word processors. It allows users to type and edit text, check spelling and grammar, format documents, and insert images, tables, and charts. It is widely used in schools, offices, and businesses.

Google Docs

Developer: Google

It is a web-based word processor that works through the internet. Multiple users can edit the same document at the same time, making it very useful for group work and collaboration.

Apple Pages

Developer: Apple

It is used on Apple devices such as MacBooks, iPhones, and iPads. It provides ready-made templates and design tools to create attractive documents easily.

LibreOffice Writer

Developer: The Document Foundation

It is a free and open-source word processor. It offers many features similar to Microsoft Word and is a good option for users who prefer free software.

Spreadsheet Software

Spreadsheet software is used to organize data in rows and columns. It helps in calculations, data analysis, budgeting, and creating charts and graphs.

Examples of Spreadsheet Software

Microsoft Excel

Developer: Microsoft

It is widely used for calculations, financial records, data analysis, and charts. It supports formulas and functions to perform complex calculations easily.

Google Sheets

Developer: Google

It is an online spreadsheet tool that allows users to work together in real time. It is useful for teamwork and can be accessed from any device with internet access.

Apple Numbers

Developer: Apple

It is designed for Apple users and is known for its clean layout and visually appealing charts. It is suitable for simple calculations and data presentation.

LibreOffice Calc

Developer: The Document Foundation

It is a free spreadsheet program used for data entry, calculations, and analysis. It provides many features similar to Microsoft Excel.

Graphic Design Software

Graphic design software is used to create, edit, and design images, posters, logos, and illustrations. It is widely used in advertising, media, and web design.

Examples of Graphic Design Software

Adobe Photoshop

Developer: Adobe

It is used for photo editing, image enhancement, and digital artwork. Designers use it to edit photos, create posters, and design graphics.

Adobe Illustrator

Developer: Adobe

It is used to create logos, icons, and illustrations. Graphics made in Illustrator can be resized without losing quality.

CorelDRAW

Developer: Corel

It is used for professional graphic design, logo creation, and page layouts. It is popular in printing and advertising industries.

GIMP[GNU Image Manipulation Program]

Developer: GIMP Development Team

It is a free and open-source image editing tool. It is used for photo editing and graphic design and is a popular alternative to Photoshop.

Canva

Developer: Canva Pty Ltd

It is an online design tool with ready-made templates. It is easy to use and is commonly used to create posters, presentations, and social media graphics.